DAMPED OSCILLATIONS AND THE THREE KINDS OF NUMBER

One spring, three kinds of number

Remove the decay: circular motion, a straight line and hyperbolic motion emerge from the same spring.

1 Start with A-level SHM

Measure x from equilibrium; $m, k > 0$ are constant.

$$m\ddot{x} = -kx, \qquad \omega_0 = \sqrt{k/m}.$$

$$x(t) = R\cos(\omega_0 t + \phi).$$

Add viscous drag from the treacle

Model the fluid by $F_d = -b\dot{x}$, with constant $b \geq 0$.

$$m\ddot{x} + b\dot{x} + kx = 0, \qquad \gamma = \frac{b}{2m}.$$

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0.$$

2 Remove the common exponential

$$x = \mathrm{e}^{-\gamma t}y \quad \Longrightarrow \quad \ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = \mathrm{e}^{-\gamma t}\big[\ddot{y} + (\omega_0^2 - \gamma^2)y\big].$$

$$\boxed{\ddot{y} = \Delta y, \qquad \Delta = \gamma^2 - \omega_0^2.}$$ The sign of Δ selects the remaining motion.

3 Three reduced motions: $y(t) = A\,C(t) + B\,S(t)$

	Underdamped complex / elliptic	**Critical** dual / parabolic	**Overdamped** split-complex / hyperbolic
condition	$\gamma < \omega_0$	$\gamma = \omega_0$	$\gamma > \omega_0$
generator	$\mathrm{i}^2 = -1$	$\varepsilon^2 = 0,\ \varepsilon \neq 0$	$\mathrm{j}^2 = +1$
$C(t)$	$\cos(\Omega t)$	1	$\cosh(\kappa t)$
$S(t)$	$\sin(\Omega t)/\Omega$	t	$\sinh(\kappa t)/\kappa$
rates r	$-\gamma \pm \mathrm{i}\Omega$	$-\gamma$ (double)	$-\gamma \pm \kappa$

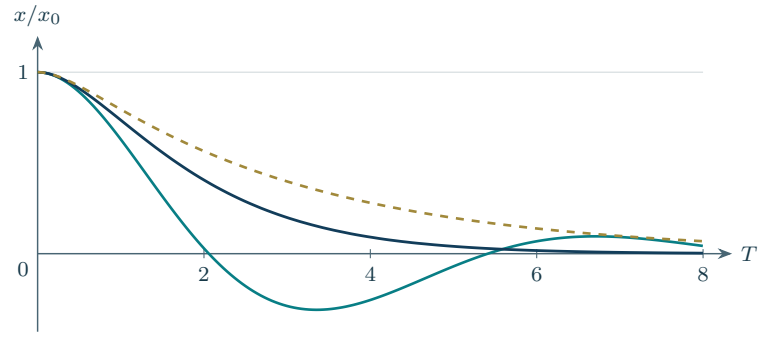
$$\Omega = \sqrt{\omega_0^2 - \gamma^2}, \qquad \kappa = \sqrt{\gamma^2 - \omega_0^2}; \qquad \gamma = \omega_0 \iff b = 2\sqrt{mk}.$$

For $x(0) = x_0$ and $\dot{x}(0) = v_0$, every regime uses the same initial-data formula:

$$\boxed{x(t) = \mathrm{e}^{-\gamma t}\big[x_0 C(t) + (v_0 + \gamma x_0)S(t)\big].}$$

Same release, different drag

Same m, k; $x(0) = x_0 > 0$, $\dot{x}(0) = 0$. Write $T = \omega_0 t$.



Damping ratio $\zeta = \gamma/\omega_0$: under 0.35, critical 1, over 1.6.

Critical: an affine factor

$$x = \mathrm{e}^{-\gamma t}(A + Bt).$$

$$\frac{\sin(\Omega t)}{\Omega} \longrightarrow t \longleftarrow \frac{\sinh(\kappa t)}{\kappa}.$$

As $\Omega, \kappa \to 0$, also $C(t) \to 1$. The reduced motion is the straight line $A + Bt$.

Overdamped still decays

Since $0 < \kappa < \gamma$, both rates below are negative:

$$\mathrm{e}^{-\gamma t}\cosh(\kappa t) = \tfrac{1}{2}\big[\mathrm{e}^{-(\gamma-\kappa)t} + \mathrm{e}^{-(\gamma+\kappa)t}\big].$$

Rotation, shear, boost: the same algebra as velocity addition

Rescale time so that $y_{\theta\theta} = \sigma y$, with $\sigma = -1, 0, +1$. Its state (y, y_θ) has generator

$$Q_\sigma = \begin{pmatrix} 0 & 1 \\ \sigma & 0 \end{pmatrix}, \qquad Q_\sigma^2 = \sigma I.$$

$$\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta, \qquad \mathrm{e}^{\varepsilon\theta} = 1 + \varepsilon\theta, \qquad \mathrm{e}^{\mathrm{j}\theta} = \cosh\theta + \mathrm{j}\sinh\theta.$$

At criticality $Q_0 \neq 0$ but $Q_0^2 = 0$, so $\mathrm{e}^{\theta Q_0} = I + \theta Q_0$: the dual-number case is a shear, and its time dependence is linear.

Physics: OpenStax, *University Physics* 1, §15.5. Geometry: V. V. Kisil (2011), *Ladder Operators in Hypercomplex Mechanics*, arXiv:1103.1120.